Data Science in Finance (DSF182)

SS3 Part A Report

Farm-Gate Price Risk and Repayment Buffers for Khula!'s Funders

A Data-Quality Review of StatsSA's Producer Price Index

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Assessment	SS3, Part A
Startup	Khula!
AI Conversation Link	https://claude.ai/share/61e93de4-793b-4396-a256-ddcdc45c32cb

Question	Response
Business question	Khula! offers farmers a single application to several funders for production finance (Khula!, 2025). If that finance is repaid from the proceeds when the crops or animals are sold, a fall in farm-gate prices between financing and sale puts repayment at risk. **Using StatsSA's monthly Producer Price Index (January 2012 – August 2026), what six-month percentage fall in farm-gate prices marks the worst 25%, 20%, 10% and 5% of six-month periods for vegetables (tomatoes, onions, potatoes), cereal grains (maize, wheat) and live animals (cattle, sheep, pigs, poultry)? Are the differences between the groups large enough to justify Khula!'s funders applying different repayment buffers at each level the data can reliably test?** *Definitions:* the fall at the Xth percentile is the six-month percentage fall reached or exceeded in X% of six-month periods (for example, the 10th percentile is reached or exceeded in one period in ten). A level can be "reliably tested" if every group has at least 5 non-overlapping six-month periods in its tail; in this data that is the 25th, 20th and 10th percentiles, while the 5th percentile is reported as indicative only. Different buffers are judged justified for two groups only if both parts of the criterion hold (notebook, Section 6): (a) *ordering*: one group's fall is larger at all four levels; and (b) *robustness*: at every reliably tested level, that ordering also holds at 3-, 6- and 12-month horizons and across non-overlapping windows.
Data source	**Source:** Statistics South Africa (StatsSA), Time series data: https://www.statssa.gov.za/?page_id=1847 **Dataset:** Producer Price Index (statistical release P0142.1), *PPI Historicals Elementary Indices (202608)*, Excel version, downloaded from the Time series data – Excel and ASCII format page on 7 October 2026 (Statistics South Africa, 2026b). **Contents:** 277 product-level price indices, monthly from January 2012 to August 2026, base December 2023 = 100, with each product's 2026 basket weight. This analysis uses 9 of the 24 agricultural products.
Coverage	**Question: Does the PPI cover the products, time period, geography and type of price that Khula!'s funders need to set repayment buffers for vegetable, cereal and livestock producers?** **Demonstrated (notebook):** The file contains 277 product indices from January 2012 to August 2026 (176 months), 24 of them agricultural products in five StatsSA subgroups (Sections 1 and 3). I analyse nine: tomatoes, onions and potatoes; maize and wheat; cattle, sheep, pigs and poultry. Together they make up 57.5% of the weight of the agriculture group: live animals 31.1%, cereal grains 17.5% and vegetables 8.9% (Section 4). Only three vegetables have a full history. Cabbage, carrots and pumpkins begin in December 2023, and the file has no index for vegetables such as

	spinach or peppers. The file has no region column, so every series is national (Section 1, “Other columns” output). **Documented:** The PPI “measures changes in the prices received by domestic producers for their outputs” (Statistics South Africa, 2026a, p. 16). For the nine products, prices come from markets (vegetables and cereal grains), auctions (cattle, sheep and pigs) and an association (poultry) (Statistics South Africa, 2026a, p. 29). A product enters the basket only if it falls in “the top 80 cumulative percentage of its group” (Statistics South Africa, 2026a, p. 12), so minor products are excluded by design. Khula!’s website describes a farmer financed to plant potatoes (Khula!, 2025), and potatoes are in the index. **Inferred:** Market and auction prices are close to, but not the same as, what a farmer receives and repays from. Any crop Khula!’s farmers grow outside the basket has no index (how far the index stands for Khula!’s own farmers is discussed under Representativeness). National series cannot show whether falls are larger in particular provinces. **Answer:** Coverage is good for cereal grains and live animals: more than 14 years of producer prices, carrying 48.6% of the agriculture group’s weight between them. It is limited for vegetables, which rest on three products and 8.9% of the weight. Funders can use the buffers for the nine products analysed, should apply the vegetable result only to tomatoes, onions and potatoes, and need other evidence for any other crop and for regional differences.
Completeness	**Question: Within the products and period used, are any monthly prices missing that could distort the bad-case falls funders would use to set a buffer?** **Demonstrated (notebook):** All nine products have a value in every one of the 176 months, giving 170 six-month changes each (Sections 3–5). Of the 24 agricultural products, 17 are complete. The other seven start late: carrots, cabbage and pumpkins in December 2023, soybeans in December 2020, pears in January 2015, and grapes and lemons in December 2013. For each, the number of months equals the span from first to last month, so these are late starts, not gaps within a series (Section 3). The file has no column or flag marking imputed values (Section 1, “Other columns” output). **Documented:** Pumpkins, carrots and cabbage are listed as new products in StatsSA’s January 2024 basket review (Statistics South Africa, 2026a, p. 13). Prices can be missing “due to the seasonal nature of the product”, and “[f]ailure to impute for missing prices may lead to a bias in the index” (Statistics South Africa, 2026a, p. 18). StatsSA fills a missing price with the average change of the other observed varieties of the product or, if none were observed, with the change in a higher-level aggregate (Statistics South Africa, 2026a, p. 18). For the nine products, Annexure B1 lists the price as a monthly average or weighted average price (Statistics South Africa,

	2026a, p. 29); such period prices "record the price across the month and so are average prices for the month" (Statistics South Africa, 2026a, p. 15) and yield "a smoother time series" (Statistics South Africa, 2026a, p. 16). **Inferred:** Some months that look complete may hold imputed prices that follow the movement of other varieties or of the wider group. Together with monthly averaging, this could make the worst falls look slightly milder than what an individual farmer experienced. **Answer:** No months are missing for the nine products, so the bad-case falls are not distorted by gaps. Leaving out the late starters keeps all groups on the same period, which means buffers cannot yet be set for crops such as cabbage. "Complete" does not mean every value was observed: some may be imputed, and the file cannot show which ones. The bad-case falls may therefore understate what individual farmers face. As a precaution rather than a finding, funders should treat them as a floor for the buffer, not an exact figure.
Consistency	**Question: Are the nine price series measured on the same basis across products and throughout January 2012 – August 2026, so that six-month falls can be compared between groups and over time when setting buffers?** **Demonstrated (notebook):** All 277 series equal exactly 100 in December 2023, so every product is on the same reference base (Section 2). The analysis compares six-month percentage changes within each product rather than index levels across products, so the choice of base does not affect the results (Section 5). The file starts in January 2012. When every six-month change that uses a 2012 value is removed, the bad-case falls change by at most 1.6 percentage points, the ordering of the groups is the same at every level, and part (b) of the criterion gives the same verdict for all three pairs (Section 7, final cells). **Documented:** "All indices were re-referenced to 100 in December 2023", and re-referencing "simply changes the level of the indices and does not affect the rates of change previously published" (Statistics South Africa, 2026a, p. 11). "New weights are introduced with effect from each January" (Statistics South Africa, 2026a, p. 22), and the latest basket review was implemented in January 2024 (Statistics South Africa, 2026a, p. 12). The PPI for agriculture, forestry and fishing was "introduced in January 2013" (Statistics South Africa, 2026a, p. 1), a year after the file begins. **Inferred:** Annual reweighting and basket reviews change how products are combined into higher-level indices. They should have little effect here, because each of the nine series is a single product's index. The report does not say how the 2012 values were produced, so they may not have been collected in the same way as later years. **Answer:** The series are consistent enough for the buffer decision. They share one reference base, the rebasing does not change

	percentage changes, and the analysis compares changes within products, not levels across them. The one documented break, the 2013 start of the agriculture PPI, moves the bad-case falls by at most 1.6 percentage points and leaves the verdict unchanged. Funders can therefore compare the groups' bad-case falls directly, but should recheck them after each January reweighting and basket review.
Timeliness	**Question: Is the PPI recent enough, and released soon enough, for Khula!'s funders to use it when setting buffers on loans made now, and how often should the buffers be updated?** **Demonstrated (notebook):** The file runs to August 2026 (last column M202608, notebook Section 1, "Month columns" output) and was downloaded on 7 October 2026 (notebook introduction, "Data" list). At that date, the latest available month was about five weeks old. The bad-case falls are calculated from more than 14 years of six-month changes (170 per product), so one additional month changes them very little. **Documented:** The PPI is "published monthly, within a month of the reference period" (Statistics South Africa, 2026a, p. 25). For the nine products the prices are monthly averages (Statistics South Africa, 2026a, p. 29), which are period prices as defined on p. 15, and with period prices "the index may be less timely when compared with point-in-time estimates, since the average cannot be calculated until the end of the period" (Statistics South Africa, 2026a, p. 16). New weights are introduced each January (Statistics South Africa, 2026a, p. 22). **Inferred:** A buffer is set when a loan is approved and is based on long-run history, so a publication lag of up to a month after the reference month ends does not weaken it. The same lag makes the PPI unsuitable for spotting a fall while a six-month loan is running: a fall in the first month of the loan would only appear once that month had ended and the index had been published, up to two months into the term, which is a third of a six-month loan. Falls since 2012 may not represent future falls, for example if growing conditions or markets change. StatsSA's file is replaced each month, so the analysed version (202608) needs to be kept for the results to be reproduced. **Answer:** The data is timely enough for setting buffers, which depends on long-run bad-case falls rather than the latest month. It is not timely enough for monitoring individual loans in real time. Funders should rerun the analysis at least once a year, after each January reweighting, and keep the file version each buffer was based on.
Representativeness	**Question: Do the PPI series represent the prices that the farmers financed through Khula! actually receive, so that group-level bad-case falls are a fair basis for each farmer's buffer?**

	Demonstrated (notebook): The groups are not uniform inside. At the 10th percentile, maize's six-month fall (−23.4%) is about twice wheat's (−11.1%), and pigs (−16.2%) and poultry (−14.8%) fall further than cattle (−9.4%) and sheep (−7.5%) (Section 6.5). Products within a group often do not move together: the average correlation of six-month changes is 0.04 for vegetables, 0.28 for cereal grains and 0.23 for live animals (Section 7, first cell). Vegetables carry 8.9% of the agriculture group's weight (Section 4). **Documented:** Indicator products "are selected based on their sales values in relation to total sales in a group" (Statistics South Africa, 2026a, p. 12), and the weights are based on "the latest available SUTs", with 2022 as the reference period (Statistics South Africa, 2026a, p. 11). StatsSA's general PPI survey sends questionnaires to companies (Statistics South Africa, 2026a, p. 15), but for the nine products the source column lists markets (vegetables and cereal grains), auctions (cattle, sheep and pigs) and an association (poultry), and no farmers are listed as a source for these nine (Statistics South Africa, 2026a, p. 29). Khula! helps farmers apply to several funders for production finance, and its site describes a farmer financed to plant potatoes (Khula!, 2025). **Inferred:** The index represents national prices in the main formal channels. A farmer financed through Khula! may sell through other channels, such as directly to buyers or locally, at prices that move differently. The weights are based on national data, with a 2022 reference period drawn from supply and use tables (Statistics South Africa, 2026a, p. 11); they say nothing about Khula!'s lending mix. Because products within a group differ and move partly independently, a farmer who grows one product faces that product's risk more than the pooled group's. **Answer:** The PPI is a reasonable but imperfect stand-in for Khula!'s farmers. It captures the general price risk of each product in formal national markets, but not an individual farmer's experience. The two-tier result holds at group level, but group results may hide product differences: at the 10th percentile, maize's bad case is closer to the vegetables' than to wheat's. Product-level tails rest on few independent periods, so this is not a tested basis for per-product buffers. Before the buffers are used for lending, funders should check them product by product, and against the prices Khula!'s own funded farmers receive if that data becomes available.
Interpretability	**Question: Can a reader at Khula! correctly tell what each row, column and value in the file means, and turn the results into a repayment buffer without misreading them?** **Demonstrated (notebook):** • *Structure:* each row is one product, described by six columns (cpcProdCode, Table, DESCR, SIC, cpcProdDesc, WEIGHTS_2026), followed by 176 month columns coded

	M201201 to M202608 (Section 1, “Other columns” and “Month columns” output). • *No dictionary:* the workbook has a single sheet with no notes or data dictionary (Section 1, final cells). • *Weights:* WEIGHTS_2026 sums to 100 within each industry table, so it is a product’s share of its own table (Section 1, final cells). • *Units:* values are index numbers, not rand prices; every series equals 100 in December 2023 (Section 2). • *Levels mislead:* tomatoes and maize have almost the same level standard deviation (24.8 and 23.8) but very different monthly changes (19.0% and 6.9%), which is why the analysis uses percentage changes (Section 5). • *Nominal:* average six-month changes of 3.5% to 8.7% include general inflation (Section 6.1). **Documented:** StatsSA defines the PPI as “[a] measure of the change in the prices of goods either as they leave their place of production or as they enter the production process” (OECD, 2008, cited in Statistics South Africa, 2026a, p. 1). The prices are transaction prices that “include all discounts or rebates given” (Statistics South Africa, 2026a, p. 16). Re-referencing “simply changes the level of the indices and does not affect the rates of change” (Statistics South Africa, 2026a, p. 11). “The reference period for the new PPI weights is 2022” (Statistics South Africa, 2026a, p. 11), and Annexure B lists the basket and weights “as at January 2026”, with SIC codes and the source and type of price for each product (Statistics South Africa, 2026a, p. 29). **Inferred:** Without the methods report, a reader could mistake an index of 84.5 for a price of R84.50, read “WEIGHTS_2026” as 2026 production when the weights refer to 2022, or treat a rise in the index as a real gain. Turning a price fall into a buffer also assumes that the farmer’s volume and costs stay the same, which a price index cannot show, so the translation below is illustrative. **Answer:** The data is interpretable for this decision once it is read with the methods report, but not on its own, because the file has no data dictionary. The analysis avoids the main traps: it uses percentage changes rather than index levels, labels results as nominal, and expresses buffers as a share of expected sales income. For example, at the 10th percentile vegetable prices fall by 31.4% or more, which leaves 68.6% of the expected price, so the illustrative buffer is **31.4% of expected sales income**, assuming volume and costs are unchanged. Funders should apply that percentage to each farmer’s own expected sales to get a rand figure.

Reference list

Khula! (2025) *Khula | Smart Farming*. Available at: https://www.khula.co.za/ (Accessed: 9 October 2026).

Statistics South Africa (2026a) *Producer price index: Sources and methods*. Report No. 01-42-01. Pretoria: Statistics South Africa. Available at: https://www.statssa.gov.za/publications/Report-01-42-01/Report-01-42-012026.pdf (Accessed: 7 October 2026).

Statistics South Africa (2026b) *PPI Historicals Elementary Indices (202608)* [Data set]. Statistical release P0142.1. Pretoria: Statistics South Africa. Available at: https://www.statssa.gov.za/?page_id=1847 (Accessed: 7 October 2026).